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Review text:

For a family $\mathcal{F}$ of graphs, let $\mathrm{EX}(n,\mathcal{F})$ and $\mathrm{SPEX}(n,\mathcal{F})$ denote the collections of $\mathcal{F}$-free graphs on n vertices having the maximum number of edges and the maximum spectral radius, respectively. For $\alpha \in [0,1)$, let $\mathrm{SPEX}_\alpha(n,\mathcal{F})$ denote the collection of $\mathcal{F}$-free graphs on n vertices maximizing the largest eigenvalue $\lambda_\alpha(G)$ of the matrix $A_\alpha(G) = \alpha D(G) + (1-\alpha)A(G)$, where $A(G)$ is the adjacency matrix and $D(G)$ is the diagonal matrix of vertex degrees of G; this matrix was introduced by Nikiforov [Appl. Anal. Discrete Math. **11** (2017), no. 1, 81–107; MR3648656]. A natural question is to decide when $\mathrm{SPEX}(n,\mathcal{F})$ and $\mathrm{EX}(n,\mathcal{F})$ coincide or intersect. They coincide for $\mathcal{F} = \{K_{r+1}\}$ by the spectral Turán theorem of Nikiforov [Linear Algebra Appl. **427** (2007), no. 2–3, 183–189; MR2351351], they are very different when $\mathcal{F}$ consists of an even cycle, and they are disjoint when $\mathcal{F}$ consists of a large enough odd wheel [Cioabă–Desai–Tait, European J. Combin. **99** (2022), Paper No. 103420; MR4308011]. Wang–Kang–Xue [J. Combin. Theory Ser. B **159** (2023), 20–41; MR4519328] showed that $\mathrm{SPEX}(n,\mathcal{F}) \subseteq \mathrm{EX}(n,\mathcal{F})$ for n sufficiently large whenever every graph in $\mathrm{EX}(n,\mathcal{F})$ is a Turán graph with $O(1)$ additional edges.

The authors consider families $\mathcal{F}$ with $\mathrm{ex}(n,\mathcal{F}) = O(n)$. For a graph G, let $\mathrm{ex}^G(n,\mathcal{F})$ denote the maximum number of edges in an $\mathcal{F}$-free graph on n vertices that contains a copy of G, and let $\mathrm{EX}^G(n,\mathcal{F})$ denote the corresponding collection of extremal graphs. Suppose that $K_{k,n-k}$ is $\mathcal{F}$-free and that $K_{k+1,m}$ is not $\mathcal{F}$-free for some fixed m. The main result says that if, for n sufficiently large, $\mathrm{EX}^{K_{k,n-k}}(n,\mathcal{F})$ contains one of the graphs $K_{k,n-k}$, $K_k + \overline{K_{n-k}}$, or $K_k + (K_2 \cup$

$\overline{K_{n-k-2}}$), then $\mathrm{SPEX}(n,\mathcal{F})$ consists of exactly that graph for n sufficiently large, where $+$ denotes the join. If instead $\mathcal{F}$ is finite and $\mathrm{EX}^{K_{k,n-k}}(n,\mathcal{F})$ contains a graph $K_k + X$ with $e(X) \le Qn + O(1)$ for some $Q < 3/4$, then $\mathrm{SPEX}(n,\mathcal{F}) \subseteq \mathrm{EX}^{K_k+\overline{K_{n-k}}}(n,\mathcal{F})$ for n sufficiently large; in this case, $e(X) = Qn + O(1)$ for some $Q \in \{0, 1/2, 2/3\}$, and X differs in a bounded number of edges from a disjoint union of a maximum number of copies of P_1, P_2, or P_3. For $Q \ge 3/4$, the authors show that every graph in $\mathrm{SPEX}(n,\mathcal{F})$ is of the form $K_k + Y$ with $\Delta(Y)$ bounded and $e(Y) \ge e(X) - O(n^{1/2})$, and they also find a condition on the degrees of X under which the same conclusion as for $Q < 3/4$ holds. These results generalize a theorem of Zhai–Yuan–You [Discrete Math. **348** (2025), no. 4, Paper No. 114351; MR4837770]. The threshold $3/4$ is sharp: the authors construct a finite family $\mathcal{F}$ for which $\mathrm{EX}(n,\mathcal{F}) = \{K_2 + (P_8 \cup \frac{n-10}{4}P_4)\}$ when $n \equiv 2 \pmod 4$ is large, whereas $K_2 + \frac{n-2}{4}K_{1,3}$ is $\mathcal{F}$-free with strictly larger spectral radius. The authors also derive a corollary relating spectral extremal graphs to connected edge-extremal graphs, allowing known results on connected extremal numbers to be used in several applications.

As applications, the authors recover many known results for paths, matchings, linear forests, star forests, long cycles, disjoint cycles, families of all graphs containing all trees of a given order, and minor-closed families, and they obtain several new ones. For instance, $\mathrm{SPEX}(n,\mathcal{F}) = \{K_{3k-1,n-3k+1}\}$ when $\mathcal{F}$ is the family of all graphs consisting of k disjoint chorded cycles, and $\mathrm{SPEX}(n,\mathcal{F}) = \{K_{k+1,n-k-1}\}$ when $\mathcal{F}$ is the family of cycles with k chords all incident to the same vertex. They also use a second moment argument to show that the proportion of labelled trees T on m vertices for which $\mathrm{SPEX}(n,T) = \{K_{k,n-k}\}$ for some k and all n sufficiently large tends to 1 as $m \to \infty$, complementing the work of Fang–Lin–Shu–Zhang [Electron. J. Combin. **31** (2024), no. 2, Paper No. 2.34; MR4746895].

The second main result relates $\mathrm{SPEX}(n,\mathcal{F})$ and $\mathrm{SPEX}_\alpha(n,\mathcal{F})$. Suppose that $\mathrm{ex}(n,\mathcal{F}) = O(n)$, or that $\mathcal{F}$ contains a bipartite graph F having a vertex v such that $F - v$ is a forest. If $\mathrm{SPEX}(n,\mathcal{F})$ contains $H = K_k + \overline{K_{n-k}}$ or $H = K_k + (K_2 \cup \overline{K_{n-k-2}})$ for n sufficiently large, then $\mathrm{SPEX}_\alpha(n,\mathcal{F}) = \{H\}$ for every $\alpha \in (0,1)$ and n sufficiently large. The case $H = K_{k,n-k}$ with $k > 1$ cannot be included, since $\lambda_\alpha(K_{1,n-1}) \ge \alpha(n-1) > \lambda_\alpha(K_{k,n-k})$ for α close to 1. Several known and new results on $\mathrm{SPEX}_\alpha(n,\mathcal{F})$ are derived from this.

Both main results are proved by analysing the Perron eigenvector of a spectral extremal graph: the authors show that there are exactly k vertices of large eigenvector entry, that these vertices are adjacent to all other vertices, and that the remaining entries are close to k/λ, after which local edge switchings that increase $x^T A x$ (resp., $x^T A_\alpha x$) determine the structure. Among the open

problems raised are whether the finiteness of $\mathcal{F}$ can be dropped in the cases $Q < 3/4$ and $Q \geq 3/4$, and to determine the supremum of the constants Q such that $\mathrm{EX}(n, F) \ni K_k + X$ with $e(X) \leq Qn + O(1)$ implies $\mathrm{SPEX}(n, F) \subseteq \mathrm{EX}(n, F)$ for a single graph F.

Some minor typos follow. In the proof of Proposition 1.2, the range of C should read $\frac{5}{32\sqrt{2}} < C < \frac{13}{32\sqrt{2}}$. On page 6542, the first displayed graph in "Certain small trees" should be $K_2 + \overline{K_{n-2}}$; in "Even cycles and consecutive cycles" the condition should read $\ell \geq 2$ and $\mathrm{SPEX}_\alpha(n, C_{2\ell+2}) = K_\ell + (K_2 \cup \overline{K_{n-\ell-2}})$; and in "Disjoint long cycles" the conclusions for ℓ even and ℓ odd are interchanged (cf. page 6540). On page 6543, $K_{1,n}$ should be $K_{1,n-1}$. Near the end of the proof of Theorem 1.1 (e), (f), $K_K + X$ should be $K_k + X$. In the proof of Lemma 3.8, $L' = k + 1$ should read $|L'| = k + 1$. References [52] and [53] list the same article.